Fiscal Year:	FY 2006	**Task Last Updated:**	FY 11/17/2006
PI Name:	Fitts, David J B.S.		
Project Title:	MSIS Reinvention		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Space human factors engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	281-483-6647
Organization Name:	NASA, Johnson Space Center		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	11/24/2002	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 9/30/2006 per B. Woolford, JSC. Project has been redefined (9/29/06)		
Key Personnel Changes/Previous PI:	Dane Russo, PhD, has taken over as PI (11/2006). See under Russo for new task project beginning October 2006.		
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
	First published in 1987, NASA-STD-3000/“Man-Systems Integration Standards” (MSIS) was prepared to promote proper integration of space human factors engineering (SHFE) requirements with those of other aerospace disciplines. Though two formal updates to the MSIS occurred in 1989 and 1995, resources have not been available to consistently keep the MSIS current. There has been a considerable amount of new spaceflight experience and SHFE research in recent years that has not been captured in the MSIS. In 2003, the Habitability and Human Factors Office (HHFO) at JSC proposes a complete revitalization of the MSIS based on new human spaceflight data from the NASA/Mir and International Space Station (ISS) programs, experience and lessons learned from MSIS integration into the ISS Program, and advancements in data presentation technologies using electronic media. Specific updates proposed for the MSIS include: * Content update based on current knowledge from spaceflight experience or research * Rationale and verification data for every requirement or standard * Consistent wording of		

Task Description:	requirements for direct inclusion into binding contractual agreements * Direct inclusion of source data * Expanded inclusion of examples from spaceflight experience * Identification of the level of maturity of each topic area's data * Information on and examples of HF integration processes * Inclusion of HF integration into all phases of a program's/project's existence—e.g., operations * Electronic presentation, hyperlinked data cross-references and Internet publication * A variety of output products and/or presentation formats tailored to various user communities Since its inception, the MSIS has been the primary tool by which the HHFO has implemented human factors research and lessons learned into practice. It is imperative to SHFE integration into NASA spaceflight projects and to HHFO core activities that the MSIS be current and relevant. If implemented, an MSIS update plan will have positive influence on the operational era of ISS.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	These new standards will serve as an anchor point by which to judge compliance with acceptable levels of risk in the "Standards to Deliverables" approach to mitigation of the crew health, safety and performance risks associated with human spaceflight.
Task Progress:	In FY06, SLSD completed development of a Spaceflight Human System Standard (SHSS) to replace NASA-STD-3000 as a new NASA standard for crew health, habitability, environmental and human factors. The SHSS is composed of 2 volumes. Volume 1, the Spaceflight Crew Health Standards (SCHS), contains medical levels of care, permissible exposure limits, fitness for duty criteria, and permissible outcome limits as a means of defining successful operating criteria for the human system. Volume 2, the Spaceflight Habitability and Human Factors Standard (SHHFS), contains environmental health and human factors standards for all human spaceflight programs. Volume 1 has been approved through the JSC SA Directorate board and is in review at NASA HQs. Volume 2 has completed JSC HEFD CCB review and will be taken next to an SA board. Note Dane Russo, PhD, has taken over as PI (11/2006). See under Russo for new task project beginning October 2006.
Bibliography Type:	Description: (Last Updated: 02/27/2004)